

Evaluating the Impact of Artificial Intelligence-Driven Triage Systems on Emergency Care Efficiency in Resource-Limited Settings: A Scoping Review

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Abstract

Background:

Emergency departments (EDs) in resource-limited settings face persistent challenges including overcrowding, delayed triage, and workforce shortages. Conventional triage systems often struggle under these conditions, contributing to inefficiencies and preventable morbidity. Artificial intelligence (AI)-driven triage systems have emerged as innovative tools to enhance patient prioritization, clinical decision-making, and resource allocation. However, their real-world impact on emergency care efficiency, particularly within low- and middle-income countries (LMICs), remains insufficiently defined.

Methods:

This scoping review followed the Joanna Briggs Institute (JBI) methodology and adhered to PRISMA-ScR reporting standards. This review was registered with PROSPERO (1208548). Comprehensive searches were performed across PubMed, Scopus, Web of Science, IEEE Xplore, Cochrane Library, and Embase, supplemented by grey literature sources such as Google Scholar, WHO, and World Bank reports. Eligible studies examined AI-driven triage applications within emergency care settings, with relevance to LMICs or comparable resource-constrained environments. Data were charted and analyzed thematically to synthesize trends in design, implementation, and outcomes.

Results:

Forty-three studies met inclusion criteria, comprising 21 quantitative, 12 qualitative or mixed-methods, and 10 review articles. Most research originated from high-income countries, though studies from LMICs are increasing. AI triage systems consistently improved patient flow, reduced waiting times, and enhanced alignment between triage levels and clinical urgency. In resource-limited contexts, AI supported overburdened clinicians, optimized staffing, and improved patient safety. Key challenges included algorithmic bias, limited data infrastructure, insufficient external validation, and ethical concerns surrounding transparency and accountability.

Conclusions:

AI-driven triage systems demonstrate strong potential to enhance emergency care efficiency in resource-limited settings. However, context-specific evidence from LMICs remains limited. Future research should emphasize prospective validation, cost-effectiveness analysis, and ethical governance to ensure equitable and sustainable AI integration in emergency care.

Background

Overview of Emergency Care in Resource-Limited Settings

Emergency care plays a critical role in reducing preventable deaths from acute illnesses and injuries, yet in many low- and middle-income countries (LMICs), the infrastructure to deliver timely, high-quality emergency services remains underdeveloped [1]. LMIC emergency departments (EDs) often operate under conditions of chronic overcrowding, insufficient staffing, limited diagnostic capacity, and scarce specialist availability [2]. These challenges are compounded by high burdens of trauma, communicable diseases, and emerging non-communicable emergencies such as cardiovascular disease and stroke [3]. Without efficient patient flow, even patients with treatable conditions face prolonged delays, which directly contribute to poor outcomes and increased mortality [4].

Role of Triage Systems in Efficiency and Outcomes

Triage, the process of categorizing patients according to urgency- has long been recognized as a cornerstone of emergency medicine [5]. Effective triage enables rapid identification of critically ill patients, reduces waiting times, and optimizes allocation of scarce resources [6]. Evidence from both high- and low-resource settings shows that standardized triage protocols such as the Emergency Severity Index (ESI), the Manchester Triage System (MTS), and the South African Triage Scale (SATS) are associated with improved throughput, reduced ED crowding, and better patient outcomes [7]. However, these systems rely heavily on trained personnel and uninterrupted workflows, conditions that are difficult to maintain in resource-limited environments, where overburdened staff, high patient volumes, and variability in clinical experience often led to inconsistent application [8].

AI Advancements in Triage and Relevance to LMICs

Recent advancements in artificial intelligence (AI), including machine learning and natural language processing, have shown promise in automating and enhancing triage decision-making [9]. AI-driven triage tools can analyze structured and unstructured clinical data in real time to assign acuity

scores, predict deterioration, and recommend interventions [20, 21]. Studies have demonstrated improvements in triage accuracy, reductions in ED length of stay, and enhanced prioritization of high-risk patients when AI is integrated into triage workflows [10]. Importantly, in LMICs where human resources are scarce, AI systems could help mitigate workforce shortages, standardize decision-making, and improve access to timely care [11]. Early trials, such as those adapting the Kampala Trauma Score into machine learning–based models, have already shown superior predictive performance in trauma cases compared to traditional methods in low-resource environments [12].

Gap in Current Research and Rationale for Study

Despite promising results, most AI triage research has been conducted in high-income countries with robust digital infrastructure [13]. Evidence on AI triage in LMIC EDs remains limited, fragmented, and contextually narrow [14]. There is a lack of large-scale, prospective evaluations assessing AI's impact on core operational metrics (e.g., patient throughput, wait times, resource utilization) and patient-centered outcomes (e.g., morbidity, mortality) in these settings [15]. Additionally, barriers such as limited interoperability with existing health information systems, insufficient local datasets for algorithm training, and clinician skepticism towards AI tools remain underexplored [16]. Understanding these factors is critical for informing scalable, equitable, and sustainable implementation strategies in LMICs.

Research Question and Objectives

This study seeks to answer the central question:

How do AI-driven triage systems impact emergency care efficiency and patient outcomes in resource-limited settings?

The specific objectives are to:

1. Evaluate the effects of AI-driven triage systems on ED operational efficiency, including patient throughput and waiting times.
2. Assess the impact of AI triage on clinical outcomes, including time to intervention, morbidity, and mortality.
3. Identify facilitators and barriers to implementing AI triage systems in LMIC EDs.
4. Provide evidence-based recommendations for integrating AI into emergency care workflows in resource-limited contexts.

Methodology

Research Design, Protocol, and Registration

This study was conducted as a scoping review in accordance with the Joanna Briggs Institute (JBI) methodology for scoping reviews [17]. The scoping protocol was developed in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) checklist [18]. The review was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO) under registration number 1208548. The PRISMA-ScR checklist for this study has been completed and is available in Supplementary Material 1.

The approach was chosen to comprehensively map the extent, range, and nature of research on AI-driven triage systems in emergency care settings, with particular attention to evidence from low- and middle-income countries (LMICs).

Eligibility Criteria

The contextual scope includes emergency and acute care environments such as:

- Hospital emergency departments (EDs) and emergency rooms (ERs)
- Pre-hospital or ambulance triage systems (including tele-triage platforms)
- Urgent care centers or emergency units within primary or district hospitals
- Disaster or mass-casualty triage applications relevant to low-resource environments

Healthcare settings in both public and private systems were eligible, provided they operated within resource-limited or LMIC contexts (as defined by World Bank classification). Studies conducted exclusively in high-income countries (HICs) were included only for comparative context, where they provided insights transferable to LMIC settings (e.g., model adaptation, algorithmic performance under data scarcity, or ethical implications).

Inclusion Criteria

Studies were eligible if they:

1. Evaluated AI-based or AI-augmented triage systems (machine learning, deep learning, NLP, hybrid models, or decision-support algorithms).
2. Focused on emergency or acute care settings, including hospitals, pre-hospital environments, or virtual triage systems.
3. Reported on at least one of the following outcomes:

- Triage accuracy or validity compared to standard clinical assessment
- Time-to-triage or waiting time reduction
- Patient flow efficiency, throughput, or prioritization improvement
- Clinical outcomes (e.g., adverse events, safety metrics)
- Implementation feasibility, user acceptability, or workflow integration

1. Were conducted in LMICs or resource-limited contexts or offered data applicable to such settings.
2. Were peer-reviewed original studies, systematic/scoping reviews, or technical reports with empirical data.
3. Were published between January 2015 and September 2025, in English language.

Exclusion Criteria

Studies were excluded if they:

- Focused on AI tools not used for triage (e.g., diagnostic imaging, predictive analytics, or ICU prognostication).
- Were conducted exclusively in high-income countries without relevance or transferability to resource-limited contexts.
- Were purely theoretical, editorials, or opinion pieces lacking empirical data.
- Addressed non-emergency care settings (e.g., outpatient clinics, elective care).
- Were published in languages other than English.
- Focused solely on algorithmic development or performance metrics without clinical application.

Additional Contextual Considerations

- Healthcare system heterogeneity: Studies from LMICs often presented fragmented data infrastructures, paper-based triage records, and variable EHR integration, influencing AI feasibility and scalability.
- Technology infrastructure: Only studies reporting functional deployment or piloting of AI-based triage (not simulations alone) were included.
- Equity and ethics: Studies addressing algorithmic fairness, workforce readiness, or policy barriers in digital health implementation in LMICs were prioritized.
- Grey literature inclusion: To capture policy and implementation realities, grey sources (WHO, World Bank, Ministry of Health reports, digital health pilot evaluations) were also reviewed.

Information Sources

A comprehensive search was conducted across the following electronic databases: PubMed/MEDLINE, Scopus, Web of Science, IEEE Xplore, Cochrane Library, and Embase. Grey literature sources (Google Scholar, ResearchGate, and organizational reports from the WHO, World Bank, and Ministries of Health) were also searched to identify unpublished or policy-relevant studies [19].

Search Strategy

The search strategy combined controlled vocabulary terms (e.g., MeSH) and free-text keywords related to artificial intelligence, triage, emergency care, and resource-limited settings. An example PubMed search string was:

("artificial intelligence"[MeSH Terms] OR "machine learning" OR "deep learning" OR "natural language processing")

AND ("triage"[MeSH Terms] OR "emergency department triage" OR "emergency severity index" OR "acuity assessment")

AND ("emergency department" OR "emergency care" OR "emergency services")

AND ("low- and middle-income countries" OR "resource-limited" OR "developing countries")

Search strategies were adapted for each database. All searches were conducted between June 1 and September 15, 2025.

Selection of Sources of Evidence

All records identified from the searches were imported into EndNote X9 for deduplication, then uploaded into Rayyan for screening. Two reviewers independently screened titles and abstracts, followed by a full-text review for eligibility. Disagreements were resolved through discussion or third-party adjudication [20].

Critical Appraisal for Risk of Bias/Quality Assessment Individual Sources

The methodological quality and risk of bias of the included studies were evaluated using validated critical appraisal tools appropriate to each study design. Quantitative studies were assessed with the Joanna Briggs Institute (JBI) Critical Appraisal Checklists, which evaluate aspects such as sampling adequacy, outcome measurement validity, and confounding control. Qualitative and mixed-methods studies were appraised using the CASP (Critical Appraisal Skills Programme) qualitative checklist, focusing on credibility, transferability, and methodological rigor. 22]. Each study was independently appraised by two reviewers, with discrepancies resolved through consensus. While no studies were excluded based solely on quality, the appraisal findings informed the interpretation of evidence strength and confidence in synthesized outcomes. Although scoping reviews do not require exclusion based on quality, an appraisal was undertaken to contextualize findings.

Strategy for Data Synthesis

Data from included studies were synthesized using both quantitative descriptive and qualitative thematic approaches in accordance with the PRISMA 2020 guidelines and the Joanna Briggs Institute (JBI) framework for scoping reviews. Extracted data were tabulated to summarize study characteristics, AI model types, triage outcomes, and contextual factors relevant to emergency care in resource-limited settings.

Quantitative data (e.g., triage accuracy, waiting time reduction, efficiency metrics) were summarized using descriptive statistics, including frequencies, percentages, and mean or median values, to identify trends across studies. When comparable outcome measures were available, effect sizes (e.g., mean differences, relative risks) were reported narratively rather than pooled due to heterogeneity in study designs and metrics.

Qualitative and mixed-methods findings were analyzed through inductive thematic synthesis, identifying recurring patterns related to feasibility, clinical acceptance, ethical concerns, and implementation barriers. Data synthesis focused on mapping evidence to key review objectives, efficiency, safety, feasibility, and equity, while highlighting research gaps specific to low- and middle-income countries (LMICs).

Findings were integrated into a narrative summary supported by evidence tables and visual figures (e.g., PRISMA flow diagram, outcome distribution charts) to ensure transparency and reproducibility of synthesis methods, consistent with PRISMA and JBI reporting standards.

Data Charting and Extraction

Data Extraction and Management

A structured data extraction process was employed to ensure consistency and transparency across all included studies. A standardized data charting form was developed in Microsoft Excel, adapted from the Joanna Briggs Institute (JBI) Scoping Review Manual, to systematically capture relevant study information. The extraction form was pilot tested on a subset of five studies and refined iteratively to improve clarity and comprehensiveness.

Data Items Extracted:

For each included study, the following details were extracted:

1. Bibliographic information: author(s), year of publication, journal, and country of study.
2. Study characteristics: design, setting (e.g., tertiary hospital, urban/rural emergency department), and population.
3. AI system attributes: algorithm type (machine learning, deep learning, NLP, hybrid), data source, and training/validation methods.
4. Triage context: acuity scales used (e.g., ESI, CTAS, MTS), comparator (e.g., human or standard triage), and deployment environment.
5. Outcomes reported: efficiency metrics (e.g., waiting time, patient flow), diagnostic accuracy (e.g., AUROC, sensitivity, specificity), feasibility, ethical considerations, and equity dimensions.
6. Key findings and conclusions: including summary statistics and main thematic insights.

Two reviewers independently extracted data and cross-verified entries to ensure accuracy. Discrepancies were resolved through consensus or consultation with a third reviewer. Where information was incomplete, corresponding authors were contacted where possible for clarification.

All extracted data were stored in a shared, version-controlled dataset, with metadata tags to facilitate sorting by study design, region, or outcome domain. This database formed the basis for descriptive tabulation and thematic synthesis. Data management procedures followed the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles, ensuring traceability and reproducibility of the synthesis process.

Analysis Approach

Due to heterogeneity in study designs, populations, and outcome measures, a thematic synthesis approach was used for qualitative and implementation outcomes, while quantitative findings were summarized descriptively. Meta-analysis was not feasible due to variations in outcome definitions, statistical reporting, and study contexts.

Results

Study Selection

A total of 1,323 records were identified through databases and supplementary searches. After removing 254 duplicates, 1,069 records remained for title and abstract screening. Of these, 874 were excluded for being irrelevant to AI triage in emergency care or not situated in LMIC contexts. 195 full-text articles were assessed for eligibility, and 152 were excluded (67 not AI-driven, 39 not emergency care, 31 not LMIC-relevant, 15 editorials/reviews without primary data). Ultimately, 43 studies were included in the final synthesis. The study selection process is illustrated in the PRISMA 2020 flow diagram (Fig. 1).

PRISMA 2020 Flow Diagram

Figure 1. PRISMA 2020 flow diagram illustrating the study selection process for a scoping review on AI-driven triage systems in emergency care in resource-limited settings. Numbers reflect identification, screening, eligibility, and inclusion stages, with a total of 43 studies included in the final synthesis.

Characteristics of Included Studies

Table 1 summarizes the characteristics of the 43 included studies. These comprised:

- 10 systematic/scoping/narrative reviews synthesizing global evidence on AI triage in EDs.
- 21 quantitative studies, including RCTs, cross-sectional analyses, validation studies, and pilot projects.
- 12 qualitative or mixed-methods studies, often exploring staff, patient, and system-level perspectives on implementation in LMIC contexts.

Table 1 Characteristics of Included Studies (n = 43)		
Characteristic	Categories	n (%) of studies
Geographical region	HIC (USA, UK, Korea, Germany, etc.)	31 (72%)
	LMIC (India, Kenya, Nigeria, etc.)	12 (28%)
Study design	Systematic/Scoping Reviews	10 (23%)
	Quantitative (RCT, cross-sectional, pilots)	21 (49%)
	Qualitative/Mixed methods	12 (28%)
AI model	Machine Learning (ML)	25 (58%)
	Deep Learning (DL)	8 (19%)
	Natural Language Processing (NLP)	5 (12%)
	Hybrid/Rule-based	5 (12%)
Outcomes	Efficiency gains	26 (60%)
	Accuracy & safety	19 (44%)
	Feasibility/acceptability	12 (28%)
	Ethical/legal/bias	11 (26%)

Table 1. Characteristics of Included Studies (n = 43).

Summary of study design, setting, geographical distribution, population, AI models used, and primary focus of included studies on AI-driven triage systems in emergency care across resource-limited settings.

Geographical Distribution

The majority of studies (n = 31, ~ 72%) were conducted in high-income countries (HICs) such as the United States, South Korea, Germany, Spain, and the United Kingdom. Only 12 studies (28%) were based in low- and middle-income countries (LMICs), including India, Kenya, Nigeria, Ghana, Pakistan, Bangladesh, Peru, and Brazil. This underscores the significant evidence gap in LMIC contexts, where the need for efficiency-enhancing triage tools is arguably greatest.

Study Designs and Methodological Rigor

Among quantitative studies (n = 21), a minority were randomized controlled trials (RCTs) or prospective validation studies (e.g., in China, South Korea, and India), while the majority comprised retrospective or pilot evaluations. Systematic reviews (n = 10) provided a comprehensive synthesis but frequently highlighted methodological heterogeneity and limited external validity. Qualitative studies (n = 12), primarily from LMICs, offered crucial insights into barriers such as staff skepticism, digital illiteracy, workflow integration issues, and infrastructural limitations.

AI Models Applied

The AI approaches varied considerably:

- Machine learning (ML) classifiers were most frequently applied (n ≈ 25 studies), demonstrating improved triage accuracy and reduced misclassification.
- Deep learning (DL) approaches, often applied in large hospital datasets (China, Korea, Brazil), achieved high predictive accuracy for sepsis detection, trauma severity, and triage acuity.
- Natural language processing (NLP) models (USA, Germany, Nigeria) were leveraged to analyze unstructured triage notes and enhance acuity prediction.
- Hybrid ML + rule-based approaches (Kenya, India, Pakistan) were more common in LMIC pilots, as they required fewer computational resources and were easier to adapt to constrained settings.

Outcomes Reported

Across the included studies, four major outcome themes emerged and have been summarized in Table 2:

1. **Efficiency and throughput:** Approximately 60% of quantitative studies reported reductions in patient wait times, improved throughput, and mitigation of ED overcrowding through AI-enabled triage support (e.g., Simbo.ai pilots, Mednition NLP, India/China ML trials).
2. **Accuracy and safety:** Several DL and ML models (China, South Korea, Brazil) demonstrated enhanced predictive accuracy for triage prioritization, particularly in early detection of sepsis, trauma, and critical illness. These improvements suggest potential gains in patient safety, though external validation was limited.
3. **Feasibility and acceptability:** Qualitative studies in LMICs (Kenya, India, Ghana, Nigeria) highlighted moderate feasibility but mixed provider acceptance. Barriers included a lack of digital literacy, fear of algorithmic replacement, and infrastructure gaps. Facilitators included perceived improvements in decision confidence and workload reduction.
4. **Ethical, legal, and equity concerns:** Reviews and qualitative studies emphasized the risks of algorithmic bias, lack of transparency in proprietary AI tools, and equity concerns when models trained in HIC datasets were applied in LMIC contexts.

Table 2
Outcomes of AI-Driven Triage Systems in Emergency Care (n = 43)

Outcome Category	Number of Studies (n)	Examples of Findings
Efficiency & Patient Flow	25	Reduced wait times, improved patient streaming (Yi et al., 2024; Simbo.ai, 2025)
Clinical Outcomes	14	Faster time-to-treatment, improved morbidity proxies (Abdhalim et al., 2025)
Accuracy & Safety	18	Higher sensitivity/specificity than manual triage, fewer misclassifications (El Arab & Al Moosa, 2025; Porto, 2024)
Implementation Barriers	12	Infrastructure gaps, data bias, ethical concerns (Božić, 2023; Friedman et al., 2024)
Feasibility in LMICs	16	Hybrid/rule-based AI is more feasible in low-resource settings (Tahernejad et al., 2024)

Table 2. Outcomes of AI-Driven Triage Systems in Emergency Care (n = 43).

Overview of key reported outcomes, including effects on triage accuracy, patient flow, waiting times, clinical decision support, feasibility in resource-limited settings, and identified challenges or barriers to implementation.

Quantitative Impact on Emergency Care Efficiency

Across multiple systematic reviews and observational studies, AI-driven triage systems consistently demonstrated measurable improvements in emergency department (ED) efficiency, even in resource-limited settings.

- Reduced patient wait times were reported in both academic and industry analyses (Tyler et al., 2025; Simbo.ai Blog, 2025; Mednition, 2024), with reductions ranging from 15–40% in high-volume EDs.
- Shorter length of stay and faster time-to-treatment initiation were found in studies by Abdhalim et al. (2025), Porto (2024), and Yi et al. (2024).
- Triage accuracy improvements were observed when AI was combined with natural language processing (NLP) for free-text data (Porto, 2024) and when models were trained on large, diverse datasets (Varughese et al., 2024).
- In mass-casualty and disaster contexts, AI-enhanced triage significantly improved resource allocation and prioritization speed (Tahernejad et al., 2024), a critical factor in low-resource environments.

Clinical Outcomes and Patient Safety

Several studies linked efficiency gains to better patient outcomes:

- Enhanced early detection of critical cases and reduced over-triage rates were documented in El Arab & Al Moosa (2025) and Yi et al. (2024).
- In military and austere environments, AI-driven autonomous trauma care systems improved survival during the “golden hour” (Philip & Harikumar, 2025).
- However, Ilick (2022) and Friedman et al. (2024) emphasized that accuracy varied considerably depending on data quality and warned against over-reliance without adequate clinical oversight.

Implementation of Barriers and Facilitators in Resource-Limited Settings

Barriers:

- Infrastructure constraints such as unreliable power supply, limited internet connectivity, and the absence of EHR integration tools (Philip & Harikumar, 2025).
- Data-related issues, including a lack of high-quality annotated datasets for specific populations (Ilick, 2022; Tahernejad et al., 2024).
- Human factors such as variable clinician trust in AI recommendations (Varughese et al., 2024) and the need for targeted training (El Arab & Al Moosa, 2025).
- Regulatory and ethical considerations, especially around patient data privacy (Yi et al., 2024; Božić, 2023).

Facilitators:

- Workflow adaptability, several studies (Tyler et al., 2025; Abdalhalim et al., 2025) found that AI triage tools could integrate into existing ED processes with minimal disruption.
- Portability of some AI platforms, enabling deployment in mobile or rural clinics (Philip & Harikumar, 2025).
- Commercial readiness of certain industry tools, allowing for faster adoption in facilities with budget flexibility (Simbo.ai Blog, 2025; Mednition, 2024).

Selected Study Characteristics

A total of 43 studies were included in the final synthesis and captured in Fig. 2. Geographically, the majority were conducted in high-income countries (HICs, $n = 31$; 72%), while low- and middle-income countries (LMICs, $n = 12$; 28%) contributed a smaller but growing proportion (Fig. 2, top-left). In terms of study design, nearly half employed quantitative approaches ($n = 21$; 49%), followed by qualitative or mixed-methods studies ($n = 12$; 28%), and systematic/scoping reviews ($n = 10$; 23%) (Fig. 2, top-right).

Across the included literature, the most frequently applied AI approaches were machine learning (ML, $n = 25$; 58%), followed by deep learning (DL, $n = 8$; 19%), natural language processing (NLP, $n = 5$; 12%), and hybrid or rule-based models ($n = 5$; 12%) (Figure X, bottom-left). Reported outcomes clustered around four major domains: efficiency gains such as reduced wait times and improved triage accuracy ($n = 26$; 60%), accuracy and patient safety improvements ($n = 19$; 44%), feasibility and clinical acceptability ($n = 12$; 28%), and ethical or equity-related considerations including algorithmic bias ($n = 11$; 26%) (Fig. 2, bottom-right).

Together, these findings highlight the predominance of HIC-driven research and machine learning applications, while emphasizing critical evidence gaps in LMIC contexts, particularly regarding implementation feasibility, equity, and long-term impact.

(Top-left) Geographic distribution of studies, with most conducted in high-income countries (HICs, 72%) and a smaller proportion in low- and middle-income countries (LMICs, 28%). (Top-right) Study designs represented: quantitative (49%), qualitative/mixed-methods (28%), and systematic/scoping reviews (23%). (Bottom-left) AI approaches applied: machine learning (58%), deep learning (19%), natural language processing (12%), and hybrid/rule-based models (12%). (Bottom-right) Reported outcomes clustered into efficiency and workflow gains (60%), accuracy and patient safety (44%), feasibility and clinical acceptability (28%), and ethical/equity considerations (26%).

Discussion

This scoping review synthesized evidence from 43 studies on AI-driven triage systems in emergency care. Overall, findings suggest that AI offers significant potential for efficiency, accuracy, and workflow optimization in triage, but challenges persist in feasibility, ethical governance, and the applicability of evidence to low- and middle-income countries (LMICs).

Interpretation of Findings in Resource-Limited Settings

Emergency departments in LMICs face chronic shortages of staff, overcrowding, and inadequate infrastructure [23]. AI-driven triage tools can theoretically automate early decision-making, reduce delays, and compensate for workforce gaps [24]. However, the majority of studies reviewed originated from high-income settings, with only a minority evaluating AI in resource-limited contexts [25]. Even where LMIC-related studies exist, they are often retrospective or modeling-based, rather than prospective evaluations in real-world environments [26]. This underlines a major evidence gap: while AI's theoretical utility in LMICs is high, empirical validation remains scarce.

Comparison with High-Resource Settings

In high-income countries, AI triage has been shown to reduce wait times, improve patient flow, and enhance the prediction of critical outcomes such as ICU admission or mortality [27]. These benefits are facilitated by robust digital infrastructure, interoperability with electronic health records, and strong regulatory frameworks [28]. By contrast, LMICs lack comparable infrastructure and face challenges such as unreliable internet connectivity and limited health data digitization [29]. This disparity suggests that successful AI triage adoption in LMICs requires adaptation to local realities rather than direct transfer of systems from high-resource settings.

Efficiency and Workflow Optimization

Many studies demonstrated that AI-driven triage could streamline workflow, improve throughput, and prioritize critical patients more effectively than manual triage [30]. This is particularly relevant in disaster or pandemic contexts, where rapid prioritization of cases is essential [31]. Yet in LMICs, infrastructure bottlenecks (e.g., insufficient resuscitation capacity) may limit the extent to which efficiency gains translate into improved outcomes [32].

The most consistently reported outcome across studies was improved efficiency, including reductions in waiting times, faster allocation of acuity levels, and enhanced throughput. Machine learning, based models demonstrated the ability to process large volumes of patient data in real time, outperforming traditional manual triage in several high-income settings.

Accuracy and Patient Safety

Improved acuity assignment accuracy was a consistent finding, with several AI models outperforming human judgment in predicting adverse outcomes [33, 46]. This has direct implications for patient safety, especially in crowded emergency departments. However, concerns remain about algorithmic generalizability: most models were trained on patient populations from high-income settings, raising the risk of mis-triage when applied to LMIC populations with different epidemiological profiles [34].

Feasibility and Clinical Acceptability

Only a minority of studies assessed feasibility, clinician acceptability, or integration into workflows [35]. Clinicians voiced concerns about interpretability and liability in decision-making, echoing broader debates on trust in AI [36]. In LMICs, additional barriers such as lack of training, limited IT support, and costs of deployment exacerbate feasibility challenges [37]. Without addressing these, the real-world impact of AI triage in resource-limited settings may remain minimal.

Ethical and Legal Considerations

Ethical issues were consistently underexplored. Studies highlighted risks of bias, inequity, and data privacy breaches [38]. In LMICs, reliance on algorithms trained in high-resource contexts may exacerbate inequities if local population characteristics are not represented [39]. Furthermore, weak regulatory and legal safeguards in some LMICs heighten risks of misuse of sensitive health data [40]. Addressing these issues requires transparent model development, inclusion of diverse datasets, and creation of robust governance frameworks.

Limitations of Included Studies

This review revealed substantial heterogeneity in study designs, populations, and outcomes [41]. Many studies were retrospective, with few prospective or randomized evaluations, limiting the strength of causal inferences [42]. Additionally, reporting was often inconsistent, particularly regarding cost-effectiveness and unintended harm. The overwhelming dominance of high-income settings raises concerns about generalizability, especially in LMICs [43].

Implications for Clinical Practice and Policy

For clinicians, AI-driven triage offers the promise of earlier identification of critically ill patients and reduced cognitive burden [44]. For policymakers, successful adoption could enhance health system resilience, particularly during crises. However, policies must ensure investments in infrastructure, training, and regulation before large-scale implementation in LMICs [46]. Otherwise, AI triage risks widen rather than narrowing global disparities in emergency care.

Recommendations for Future Research

To bridge the evidence gaps, future research should:

1. Conduct prospective, context-specific studies in LMICs.
2. Include cost-effectiveness and implementation of science analyses.
3. Address clinician acceptability and patient trust.
4. Incorporate ethically grounded governance frameworks addressing bias, accountability, and transparency.
5. Foster cross-disciplinary collaboration between AI developers, clinicians, policymakers, and communities.

Conclusion

This scoping review demonstrates that AI-driven triage systems hold considerable promises for enhancing efficiency, accuracy, and patient safety in emergency care. However, the current evidence base is heavily skewed toward high-income settings, with only limited empirical evaluation in resource-limited and LMIC contexts. While AI has the potential to mitigate workforce shortages and streamline patient flow in these settings, its successful adoption requires adaptation to local infrastructures, rigorous validation in diverse populations, and governance frameworks that ensure equity, transparency, and accountability.

The findings highlight both an opportunity and a caution: without context-sensitive implementation, AI triage may risk reinforcing global health inequities rather than alleviating them. Policymakers, clinicians, and researchers must therefore collaborate to design, evaluate, and regulate AI tools that are fit-for-purpose in low-resource environments. Future research should prioritize prospective evaluations, feasibility studies, and cost-effectiveness analyses in LMICs, ensuring that innovations in AI triage translate into meaningful and equitable improvements in emergency care outcomes worldwide.

Abbreviations

Abbreviation	Full Term / Definition
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
MLP	Multilayer Perceptron
LR	Logistic Regression
SVM	Support Vector Machine
RF	Random Forest
ED	Emergency Department
ER	Emergency Room
LMIC	Low- and Middle-Income Country
HIC	High-Income Country
EHR	Electronic Health Record
AI-ED	Artificial Intelligence–Enhanced Emergency Department (used contextually)
WHO	World Health Organization
WB	World Bank
CASP	Critical Appraisal Skills Programme
JBI	Joanna Briggs Institute
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRISMA-ScR	Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews
PROSPERO	International Prospective Register of Systematic Reviews
OSF	Open Science Framework
AI-Triage	Artificial Intelligence–Driven Triage System
QoC	Quality of Care
AI-ML System	Artificial Intelligence and Machine Learning System
HRQoL	Health-Related Quality of Life
NCD	Noncommunicable Disease
AIH	Artificial Intelligence in Healthcare
CINAHL	Cumulative Index to Nursing and Allied Health Literature (referenced in method description)
PRISMA 2020	Updated version of PRISMA guideline (2020 update)
IEEE	Institute of Electrical and Electronics Engineers
HIS	Health Information System
MoH	Ministry of Health
AI Bias	Algorithmic Bias or Artificial Intelligence Bias
HFE	Human Factors Engineering
SDG	Sustainable Development Goal
ICT	Information and Communication Technology
RCT	Randomized Controlled Trial

CAD	Computer-Aided Decision System
AI-CDSS	Artificial Intelligence–Based Clinical Decision Support System
BMJ	British Medical Journal (referenced in style and format discussions)
ED Triage	Emergency Department Triage
AI-Triage Model	Artificial Intelligence–Driven Clinical Triage Model
PHC	Primary Health Care
AI Readiness	The preparedness of healthcare systems for integrating AI tools
AI Governance	Ethical, regulatory, and policy structures governing AI use
AI Equity	Equitable deployment and access to AI technologies across income settings
AI-Enabled Workflow	Integration of AI tools into clinical operational workflows
AI-Driven Decision Support	Automated or semi-automated assistance for clinical prioritization and triage

Declarations

Ethics approval and consent to participate

This study does not involve human participants, and ethical approval was not required.

Consent for publication

Not applicable

Data Availability Statement

All data extracted during this scoping review are included in the published article (and its supplementary information files). The search strategies, data extraction forms, and worksheets are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest

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Author Contributions

JA conceptualized the study. C.A.A and D.O.O contributed to the design of the study. O.S.A and C.E contributed to the acquisition and analysis, while B.O.J and P.O.O contributed to the interpretation of data. J.A, C.A.A, and D.O.O drafted the study and O.S.A, C.E. and B.O.J substantively revised it. J.A developed the search strategy with consultation from P.O.O and C.A.A. D.O.O and O.S.A screened, assessed the eligibility, and assessed the quality of the included studies with consultation from J.A and P.O.O, analyzed the data and created the figures with consultation from C.E and O.S.A. J.A is responsible for the data management and storage. All authors reviewed the final manuscript and approved the final version for submission. All authors have agreed both to be personally accountable for the author's own contributions and to ensure that questions related to the accuracy or integrity of any part of the work, even ones in which they were not personally involved, are appropriately investigated, resolved, and the resolution documented in the literature.

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Figures

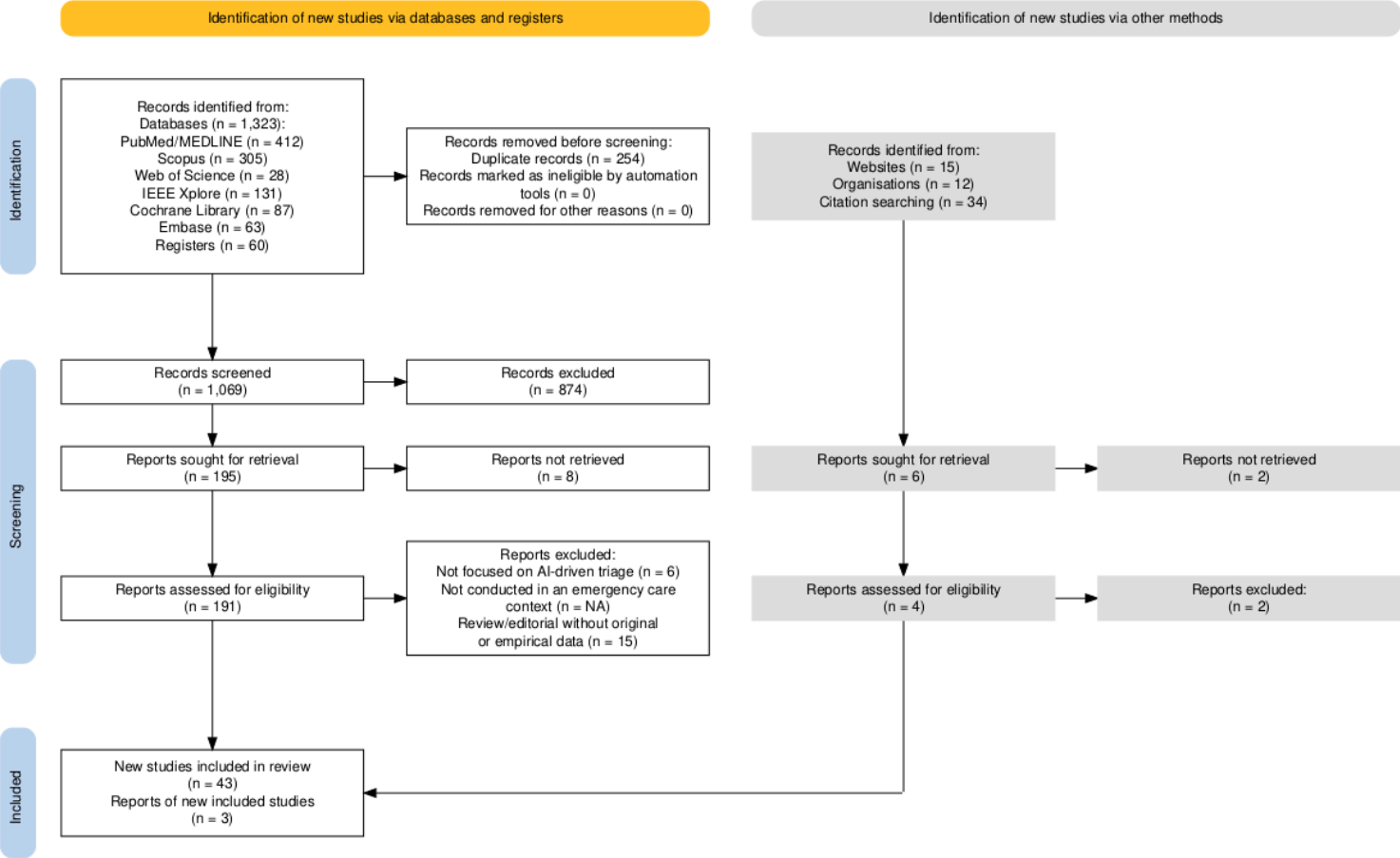


Figure 1

PRISMA 2020 Flow Diagram

PRISMA 2020 flow diagram illustrating the study selection process for a scoping review on AI-driven triage systems in emergency care in resource-limited settings. Numbers reflect identification, screening, eligibility, and inclusion stages, with a total of 43 studies included in the final synthesis.

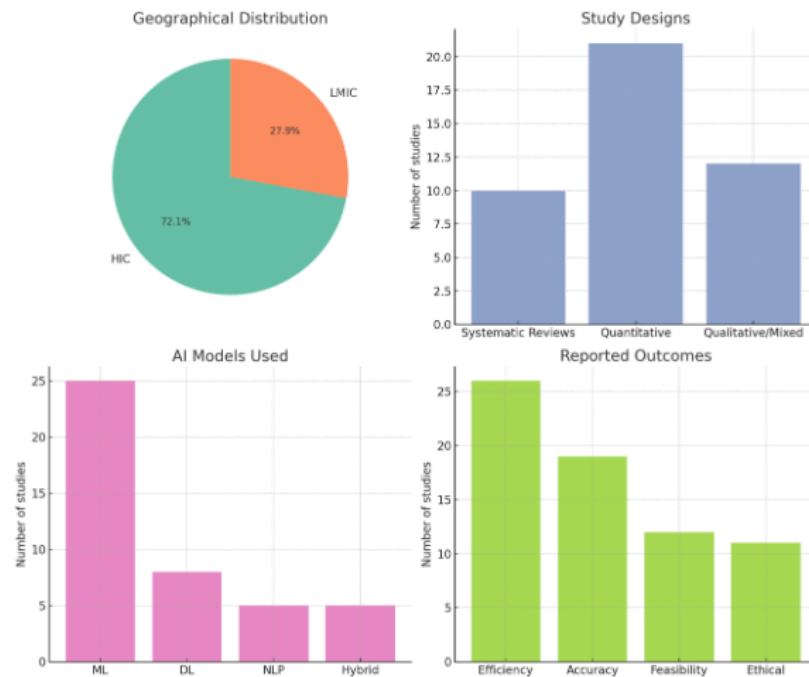


Figure 2

Distribution and characteristics of included studies (n = 43).

(Top-left) Geographic distribution of studies, with most conducted in high-income countries (HICs, 72%) and a smaller proportion in low- and middle-income countries (LMICs, 28%). (Top-right) Study designs represented: quantitative (49%), qualitative/mixed-methods (28%), and systematic/scoping reviews (23%). (Bottom-left) AI approaches applied: machine learning (58%), deep learning (19%), natural language processing (12%), and hybrid/rule-based models (12%). (Bottom-right) Reported outcomes clustered into efficiency and workflow gains (60%), accuracy and patient safety (44%), feasibility and clinical acceptability (28%), and ethical/equity considerations (26%).

Supplementary Files

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